

|    |                                                                                                                                            |
|----|--------------------------------------------------------------------------------------------------------------------------------------------|
|    | By definition for thermodynamic temperature scale.                                                                                         |
| 12 | <b>Ans: B</b><br>$q = \Delta U + w$<br>$\Delta U = q - w$ compare with $\Delta U = Q + W$ (usual convention)<br>$W$ is work done by system |
| 13 | <b>Ans: D</b>                                                                                                                              |